

BINARY SEARCH RECURSION

BINARY SEARCH RECURSION EXPLAINED

- The binary search algorithm can be implemented using iteration or recursion.
- An iterative method involves traversing through each element of the array.
- The binary search recursive method calls itself again and again until the base condition (stopping condition) is satisfied.
- The base condition is satisfied when low is greater than high, that is the space has been searched without matching the search key.

BINARY SEARCH RECURSION IMPLEMENTATION

```
public static int binarySearch(int[] data, int key, int low, int high){  
    if(low > high)  
        return -1;  
    int middle = ((low + high) + 1)/2;  
    if(key == data[middle])  
        return middle;  
    else if(key < data[middle]){  
        high = middle-1;  
        return binarySearch(data, key, low, high);  
    }  
    else{  
        low = middle+1;  
        return binarySearch(data, key, low, high);  
    }  
}
```

BINARY SEARCH RECURSION IMPLEMENTATION

- The `binarySearch` method performs the binary search.
- Four parameters - the array to be searched, the search key and the lower and upper bounds of the array.
- The base condition, is $low > high$ is first checked, if so return false (-1)
- The middle index is calculated.
- The search key is compared to the middle element of the array. If it is a match the index is returned.
- If the search key is less than the middle element, high is given a value of $middle - 1$, the method calls itself again.
- If the search key is greater than the middle element, low is given a value of $middle + 1$, the method calls itself again.
- The method continues to call itself until the search key is found or $low > high$